Q. A random sample of 500 pineapples was taken from a large consignment and 65 out of them were found to be bad. Show that the standard error of proportion of the bad ones in the sample of this size is 0.015 and deduce that the percentage of bad pineapples in the consignment almost certainly lies between 8.5 and 17.5. Also obtain 95% confidence interval for population proportion p. Construct 95% confidence interval for p if no estimate of p is available.

Soln:- Given $n = 500$

$p = \frac{65}{500}$ $\left\{ \begin{array}{l} \text{proportion of bad pineapples} \\ \text{in the sample} \end{array} \right.$

$p = 0.13$

$q = 0.87$

Standard error $SE(p) = \sqrt{\frac{pq}{n}}$

$= \sqrt{\frac{0.13 \times 0.87}{500}} = \underline{\underline{0.0150}}$

$SE(p) = 0.015$,

Hence we proved that standard error of proportion of bad pineapples in the sample of 500 size is $\underline{\underline{0.015}}$

Next we have to show that percentage of bad pineapples in the consignment almost certainly lies between 8.5 and 17.5.

For that, we know that, confidence interval for proportion is $p \pm Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}$

given $p - Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} = 8.5\%$

$p + Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} = 17.5\%$

ie $p - Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} = \frac{8.5}{100} \Rightarrow 0.13 - Z_{\alpha/2} \cdot 0.0150 = 0.085$ — (1)

$p + Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}} = \frac{17.5}{100} \Rightarrow 0.13 + Z_{\alpha/2} \cdot 0.0150 = 0.175$ — (2)

(2) − (1) $\Rightarrow$

$$\begin{aligned} &0.13 + Z_{\alpha/2} \cdot 0.0150 = 0.175 \\ -&[0.13 - Z_{\alpha/2} \cdot 0.0150 = 0.085] \\ \hline &Z_{\alpha/2} \cdot 0.03 = 0.09 \end{aligned}$$

$$Z_{\alpha/2} = \frac{0.09}{0.03} = \underline{\underline{3}}$$

$Z_{\alpha/2} = 3$ is a valid value for Z.

∴ we can say that the percentage of bad pineapples can lie between 8.5 & 17.5.

Hence proved.

To construct 95% C·I for p if no estimate is given.

$$100(1-\alpha)\% = 95\%$$

$$\alpha = 0.05 \ \& \ \alpha/2 = 0.025$$

$$\therefore Z_{\alpha/2} = 1.96.$$

If no estimate of population proportion 'p' is available, in such situations $p = 0.5$ & $q = 0.5$ for standard error.

ie, Confidence interval for p is given by

$$p \pm Z_{\alpha/2} \cdot \sqrt{\frac{pq}{n}}$$

ie $SE(p) = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.5 \times 0.5}{500}} = 0.0223$

C·I is $p \pm Z_{\alpha/2} \cdot SE(p)$

ie $0.13 \pm 1.96 (0.0223)$

0.13 ± 0.0438

ie $0.13 - 1.96(0.0223) \leq (\mu_1 - \mu_2) \leq 0.13 + 1.96(0.0223)$

ie, $0.0862 \leq (\mu_1 - \mu_2) \leq 0.1738$

C·I for μ, Large Sample

Q:. A sample of 100 gave a mean of 7.4 kg and a standard deviation of 1.2 kg. Find 95% confidence limit for the population mean.

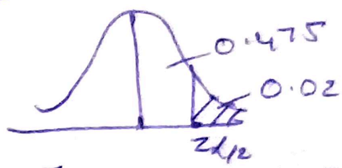
Given $n = 100$, $\bar{x} = 7.4$, $S = 1.2$

$$100(1-\alpha)\% = 95\%$$

$$1-\alpha = .95$$

$$\alpha = 1 - 0.95 = 0.05$$

$$\alpha/2 = 0.025$$



$\therefore$ $Z_{\alpha/2} = 1.96$ From tables

CI on population mean is

$$\bar{x} - Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \le \mu \le \bar{x} + Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

$$7.4 - 1.96\frac{(1.2)}{\sqrt{100}} \le \mu \le 7.4 + 1.96\frac{(1.2)}{\sqrt{100}}$$

$$7.1648 \le \mu \le 7.6352$$